

Summary

- Setup Environment for React
- Install Node JS & NPM [Yarn, Composer, RubyGEMS, NuGET]
- Install VS Code
- Install Plugins

React in existing web application:

1. Create a new web application & setup development environment

- Create a new folder for project on your PC

D:\web-app

- Open project folder in VS code

- Open Terminal in project location [ctrl + ` ] / [ctrl + shift + `]

> npm init -y

[Generates package.json file]

- Add "README.MD", It is a help document by developer for developer.

- Add ".gitignore", It is used to configure the resources to ignore while publishing on to GIT.

- Add folders

a) public	: To keep all static
resources like html, images, docs, videos etc	
b) src	: To keep all dynamic
resources like css, js, jsx, ts, tsx, sass etc	

- Add "index.html", It is the startup page for project.

web-app

├─package.json
└─readme.md

```

l_.gitignore
l_public
l_src
l_index.html

```

- Go to index.html and add `<noscript>` element.

```

<body>
  <noscript>
    please enable JavaScript on
your browser.
  </noscript>
</body>

```

- In SPA environment page will have only one container to handle everything.

```
<div id="root"> </div>
```

- To enable React in any page you need following libraries

library.	a) React	: It is the core React
manipulate virtual DOM.	b) React DOM	: It is used to create &
compiler.	c) Babel	: It is a Javascript

CDN Library for React upto 17x version

1. Visit <https://legacy.reactjs.org/>
2. Go to "DOCS => CDN Links"
3. Copy and paste development links into your HTML page.
[index.html]
4. Visit <https://babeljs.io/>
5. Go to "DOCS => Integration Packages => babel@standalone"

6. Copy and paste babel library into your HTML page.

```
react.development.js  
react-dom.development.js  
babel.js
```

7. Add a new `<script>` element and create virtual DOM, which renders content into actual DOM.

```
<script type="text/babel">  
  
</script>
```

Creating Virtual DOM

a) Upto version 17x

```
ReactDOM.render("your_message",  
document.getElementById("root"));
```

b) From version 18x upto 19x

```
const root =  
ReactDOM.createRoot(document.getElementById("root"));  
root.render("Welcome to React 18x &  
19x");
```

Ex:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-  
width, initial-scale=1.0">  
  <title>Index</title>  
  <script crossorigin src="https://unpkg.com/  
react@18/umd/react.development.js"></script>  
  <script crossorigin src="https://unpkg.com/react-  
dom@18/umd/react-dom.development.js"></script>  
  <script src="https://unpkg.com/@babel/standalone/  
babel.min.js"></script>
```

```

    <script type="text/babel">
      const root =
ReactDOM.createRoot(document.getElementById("root"));
      root.render("Welcome to React 18x & 19x");
    </script>
  </head>
  <body>
    <noscript>
      Please enable Javascript on your browser
    </noscript>
    <div id="root"></div>
  </body>
</html>

```

Install React & Babel Library

1. Run the following commands from terminal

```

> npm install react@18 react-dom@18 --save
> npm install @babel/standalone --save

```

2. Add downloaded library files are kept in “**node_modules**” folder

3. Node JS on your device uses “UMD” modules [Universal Module Distribution].

Hence you have to link library from UMD.

Syntax:

```

<script src="./node_modules/react/umd/
react.development.js"> </script>

```

Ex:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Index</title>
  <script src="./node_modules/react/umd/
react.development.js"></script>
  <script src="./node_modules/react-dom/umd/react-
dom.development.js"></script>

```

```
    <script src="./node_modules/@babel/standalone/
babel.js"></script>
    <script type="text/babel">
        const root =
ReactDOM.createRoot(document.getElementById("root"));
        root.render("Welcome to React 18x & 19x");
    </script>
</head>
<body>
    <noscript>
        Please enable Javascript on your browser
    </noscript>
    <div id="root"></div>
</body>
</html>
```

